

Class 10: Structural Bioinformatics

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PDB statistics

The Protein Data Bank (PDB) is the main repository of biomolecular structures. Let's see what it contains:

```
stats <- read.csv("Data Export Summary.csv")
stats
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178,795	21,825	12,773	343	226
2	Protein/Oligosaccharide	10,363	3,564	34	8	11
3	Protein/NA	9,106	6,335	287	24	7
4	Nucleic acid (only)	3,132	221	1,566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					
1	84 32	214,078				
2	1 0	13,981				
3	0 0	15,759				
4	3 1	4,941				
5	0 0	237				
6	0 4	22				

```
sum(stats$Neutron)
```

```
[1] 88
```

The comma in these numbers leads to the numbers here being read as character.

```
c(100,10)
```

```
[1] 100 10
```

```
library(readr)
stats <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Molecular Type
```

```
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
stats
```

```
# A tibble: 6 x 9
```

`Molecular Type` <chr>	`X-ray` <dbl>	EM <dbl>	NMR <dbl>	Integrative <dbl>	`Multiple methods` <dbl>	Neutron <dbl>
1 Protein (only)	178795	21825	12773	343	226	84
2 Protein/Oligosacch~	10363	3564	34	8	11	1
3 Protein/NA	9106	6335	287	24	7	0
4 Nucleic acid (only)	3132	221	1566	3	15	3
5 Other	175	25	33	4	0	0
6 Oligosaccharide (o~	11	0	6	0	1	0

```
# i 2 more variables: Other <dbl>, Total <dbl>
```

```
sum(stats$`X-ray`)
```

```
[1] 201582
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
total_structures <- sum(stats$Total)
xray_percent <- sum(stats$`X-ray`) / total_structures * 100
em_percent <- sum(stats$EM) / total_structures * 100
xray_percent
```

```
[1] 80.95077
```

```
em_percent
```

```
[1] 12.83843
```

Q2: What proportion of structures in the PDB are protein?

```
protein_percent <- stats$Total[stats$`Molecular Type`== "Protein (only)"] /  
sum(stats$Total) * 100  
protein_percent
```

```
[1] 85.96889
```

Visualizing the HIV-1 protease structure

We can use the Molsat viewer online: <https://molstar.org/viewer/>

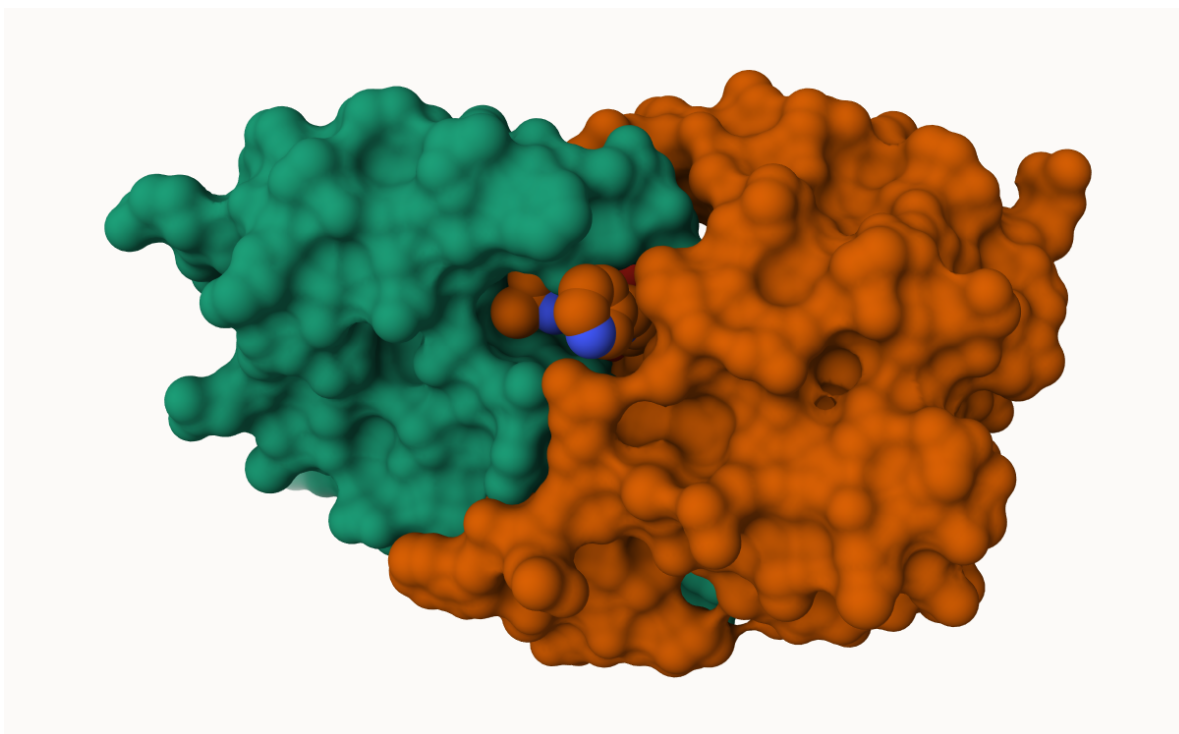


Figure 1: My first image of HIV-Pr with surface display showing ligand binding

A new clean image showing the catalytic ASP25 amino acids in both chains of the HIV-PR dimer along with the inhibitor and the all important active site water.

Bio3D package for structural bioinformatics

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

Call: read.pdb(file = "1hsg")

Total Models#: 1

Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)

Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

```
head(pdb$atom)
```

	type	eleno	ety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

```
options(repos = c(CRAN = "https://cran.rstudio.com/"))
```

```
install.packages("pak")
```

The downloaded binary packages are in
 /var/folders/r6/txgsh4vn1mv5vx1b0tgmyw6c0000gn/T//Rtmp1CAuZS/downloaded_packages

```
pak::pak("bioboot/bio3dview")
```

! Using bundled GitHub PAT. Please add your own PAT using `gitcreds::gitcreds\_set()`.

Loading metadata database

Loading metadata database ... done

No downloads are needed

1 pkg + 40 deps: kept 37 [5.2s]

```
install.packages("NGLVieweR")
```

The downloaded binary packages are in
 /var/folders/r6/txgsh4vn1mv5vx1b0tgmyw6c0000gn/T//Rtmp1CAuZS/downloaded_packages

```
library(bio3dview)
```

Predicting functional motions of a single structure

Read on ADK structure from the PDB database

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

Call: read.pdb(file = "6s36")

Total Models#: 1

Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)

Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

Protein sequence:

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM  
TAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

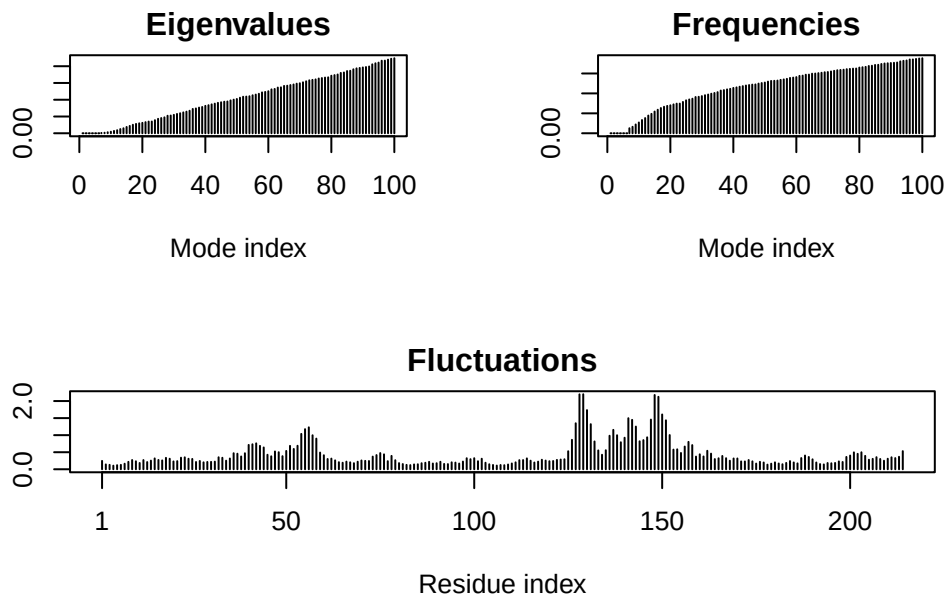
+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

```
m <- nma(adk)
```

Building Hessian... Done in 0.015 seconds.

Diagonalizing Hessian... Done in 0.061 seconds.

```
plot(m)
```



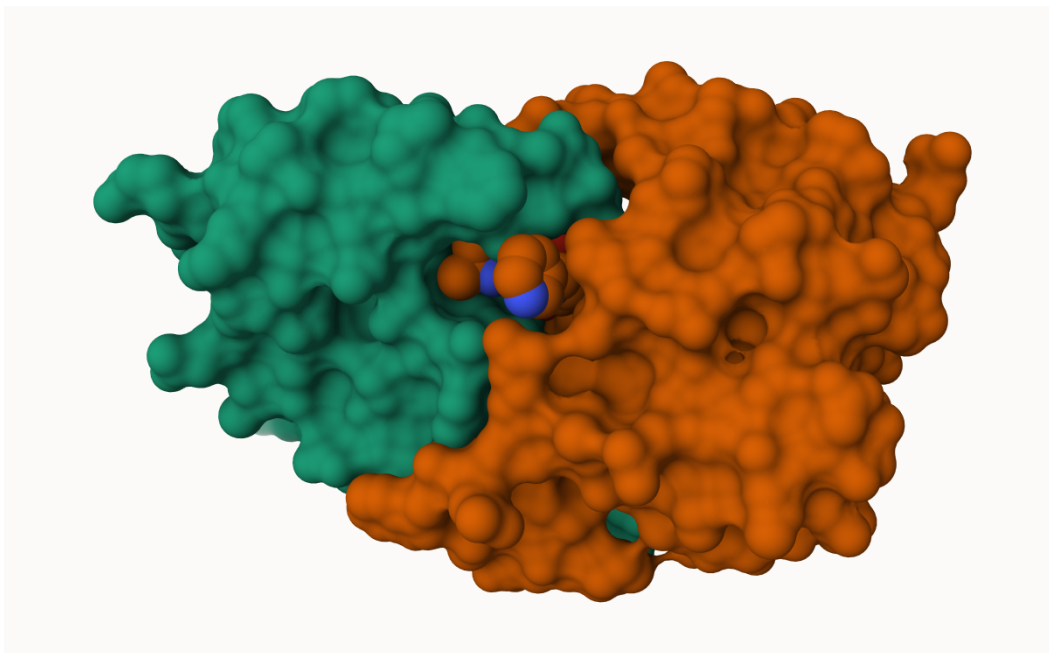
write our results as a wee trajectory/movie of a predicted motions:

```
mktrj(m, file="adk_m7.pdb")
```

```
install.packages("png")
```

The downloaded binary packages are in
/var/folders/r6/txgsh4vn1mv5vx1b0tgmyw6c0000gn/T//Rtmp1CAuZS/downloaded_packages

```
library(png)  
library(grid)  
img <- readPNG("2HSG.png")  
grid.raster(img)
```



Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Hydrogen atoms are too small to produce strong electron density in X-ray crystallography, so they usually aren't visible in the resulting structure. Because of this, PDB files typically record only the oxygen atom for each water molecule.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

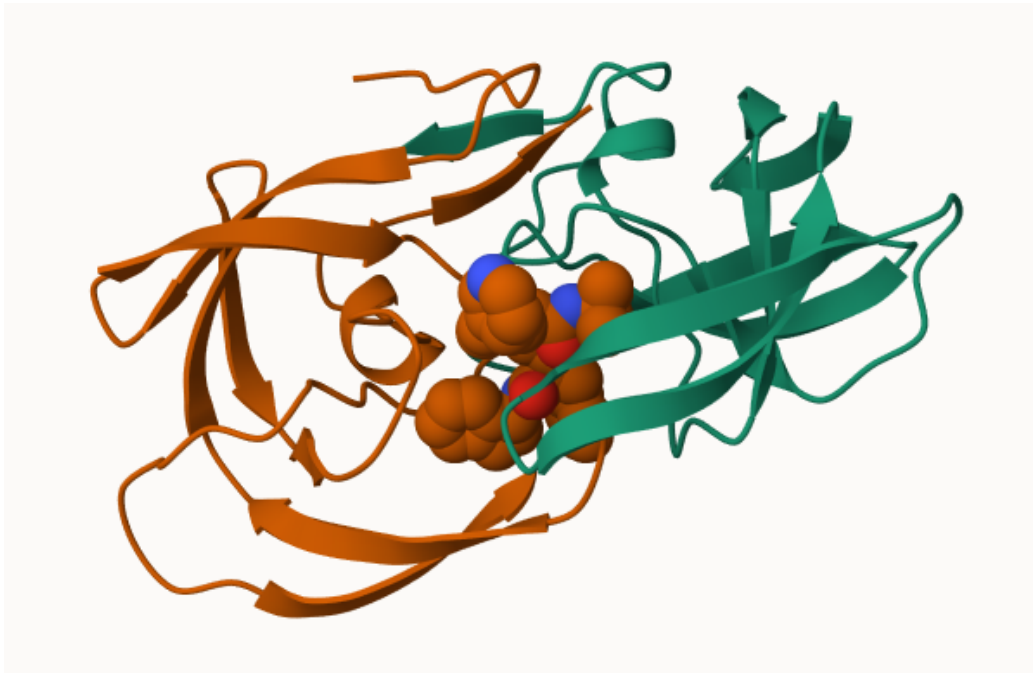
The residue number that the water molecule have is 308.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

```
install.packages("png")
```

The downloaded binary packages are in
/var/folders/r6/txgsh4vn1mv5vx1b0tgmyw6c0000gn/T//Rtmp1CAuZS/downloaded_packages


```
library(png)
library(grid)
img <- readPNG("1HSG.png")
grid.raster(img)
```



```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/r6/txgsh4vn1mv5vx1b0tgmyw6c0000gn/T//Rtmp1CAuZS/1hsg.pdb exists.
Skipping download

```
pdb
```

Call: read.pdb(file = "1hsg")

Total Models#: 1

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QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

MK1 (indinavir)

Q9: How many protein chains are in this structure?

There are 2 protein chains in this structure (chains A and B)